Balaji Radhakrishnan Padmanabhan

ASU Graduate Student | Graduating May 2027 | Tempe, AZ | +1 602-725-2850 | balajirp01121@gmail.com
linkedin.com/in/balajirpadmanabhan021 | bradhak5-asu.github.io/Balaji-Portfolio | github.com/bradhak5-ASU

Education

Arizona State University, Ira A. Fulton Schools of Engineering — Tempe, AZ
Master of Science in Software Engineering | GPA: 3.56/4.00 — *Aug 2025 – Expected May 2027*

Relevant Coursework: Advanced Data Structures and Algorithms, Data Processing at Scale, Web Development

SRM Institute of Science and Technology — Chennai, India
Bachelor of Technology in Computer Science and Engineering | CGPA: 8.64/10.00 — *Jun 2018 – May 2022*

Technical Skills

Languages: Python, SQL, JavaScript, TypeScript, Bash, PowerShell
Backend & Data: FastAPI, Flask, SQLAlchemy, Alembic, Pydantic, REST APIs, JWT, PostgreSQL, MySQL, Pandas
Cloud & DevOps: Google Cloud (Cloud Run, Cloud SQL, Cloud Build, Cloud Scheduler, Secret Manager), AWS (S3, RDS), Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, Git, Linux, TCP/IP
Testing & Web: React, Vite, Pytest, Playwright, Selenium, Postman

Experience

Arizona State University – Enterprise Technology, MIX Center — Mesa, AZ
IT Deskside Support Assistant (Part-time) — *Mar 2026 – Present*

- Resolve 10+ weekly ServiceNow incidents, troubleshoot endpoint TCP/IP connectivity, and automate configuration and deployment validation with PowerShell, Bash, and Linux/macOS tooling across fleets managed with SCCM, Intune, and Jamf.

Tata Consultancy Services | Client: Cigna Healthcare — Chennai, India
Systems Engineer, Automation (promoted from Assistant Systems Engineer) — *Sep 2022 – Jul 2025*

- Developed and extended Python, Pytest, and Selenium automation tooling for a healthcare RFP platform, integrating execution and reporting into Jenkins CI/CD across 50+ Agile sprints.
- Built SQL and Pandas validation checks for PostgreSQL and AWS S3 data pipelines, catching nulls, duplicates, schema drift, and transformation mismatches between source and target.
- Debugged distributed workflows spanning Salesforce, Pega, a Next.js portal, REST APIs, and downstream ETL systems, tracing payloads, logs, and transformations to isolate defects at system boundaries.

Projects

TriageZero: Agentic Release Engineering Platform | *Python, FastAPI, Google ADK, Vertex AI, Cloud Run* — 2026

Google All Things Agentic Hackathon | github.com/bradhak5-ASU/TriageZero-AI

- Built and deployed a FastAPI backend and React dashboard on Google Cloud that ingest regression-failure evidence through a versioned REST contract and return an evidence-backed root cause, severity, release risk, and recommended action; classified all 84 evaluation cases correctly within the supported defect class.
- Orchestrated a seven-stage Google ADK workflow calling Gemini through Vertex AI with read-only evidence tools, validating output against a closed schema with a deterministic fallback analyzer and keeping release-risk policy outside the model.
- Deployed four Cloud Run services and two scheduled jobs on Cloud SQL, automated builds through Cloud Build, and hardened ingestion with token authentication, schema validation, sanitization, idempotency-key deduplication, and telemetry.

Inventory and Equipment Lending API | *FastAPI, PostgreSQL, SQLAlchemy, Alembic, Docker, Pytest* — 2026

github.com/bradhak5-ASU/inventory-management-api

- Designed a normalized eight-table schema for a multi-role lending service covering consumables and serialized assets.
- Modeled stock as an append-only movement ledger with database-enforced invariants: check constraints, row-level locking, and a partial unique index guaranteeing one active loan per unit; secured endpoints with JWT and a role-gate dependency.

Global Superstore Data Pipeline | *Python, PostgreSQL, SQL, Docker, Tableau, Gemini API* — 2026

github.com/bradhak5-ASU/Global-SuperStore-Data

- Built an idempotent, transactional ELT pipeline that loaded 51,290 retail records into PostgreSQL and ran transformations in-database, so reruns never duplicated or corrupted rows.
- Reconstructed all 51,290 corrupted source dates to week-level accuracy with zero mismatches against order-ID years, then exposed governed SQL views for Tableau and a Gemini question-answering layer.

Publication

Designing a Dynamic Framework for People Counting Using YOLO-PC | AIP Conf. Proceedings, 2024 | ICCCN 2022